Class 6 Science – Chapter 11: Nature’s Treasures

Complete Question–Answer Notes in Simple English | Textbook pages 153–160

इस अध्याय में वायु, जल, सूर्य, वन, मिट्टी, चट्टानें, खनिज, जीवाश्म ईंधन, प्राकृतिक संसाधनों का संरक्षण और सतत विकास के बारे में बताया गया है। नीचे पुस्तक के पृष्ठों पर दिए गए प्रश्नों और गतिविधियों के उत्तर के साथ प्रत्येक टॉपिक के महत्वपूर्ण संभावित प्रश्न-उत्तर दिए गए हैं।

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1. Air Is Everywhere (Page 153)

Q1. What do you see in a forest or garden?

Answer: I see trees, plants, flowers, birds, insects, and other living things.

Q2. Name some things that we get from forests.

Answer: We get wood, fruits, medicines, honey, gum, oxygen, and many other useful products from forests.

Q3. What do you feel when you close your eyes on a windy day?

Answer: I can feel the cool air moving around my face and body.

Q4. How do you know that air is around you?

Answer: We can feel moving air on our skin. We can also see leaves moving, flags waving, and kites flying because of air.

Q5. Does the air around you have any smell?

Answer: Clean air has no smell. However, air may have a smell when it contains smoke, perfume, or other substances.

Q6. Can you see air?

Answer: No, we cannot normally see air because it is colourless and transparent. We can feel it when it moves.

Q7. Does air have any size or shape?

Answer: Air has no fixed shape of its own. It takes the shape of the container in which it is present. Air also occupies space.

2. Resources and Their Types (Page 154)

Q1. What are resources?

Answer: Resources are materials or things that are useful to humans and help meet their needs.

Q2. What are natural resources?

Answer: Natural resources are materials obtained from nature, such as air, water, soil, forests, sunlight, rocks, and minerals.

Q3. What are human-made resources?

Answer: Human-made resources are things made or modified by humans using natural materials and technology.

Examples: Buildings, roads, bridges, furniture, electric bulbs, and solar panels.

Q4. What are human resources?

Answer: Human resources refer to people and their knowledge, skills, and abilities that help produce goods and provide services.

Q5. What are the main types of natural resources based on renewability?

Answer: The two main types are renewable resources and non-renewable resources.

Q6. Differentiate between natural and human-made resources.

Natural resources	Human-made resources
Obtained from nature.	Made or modified by humans.
Examples: Air, water, sunlight, and minerals.	Examples: Roads, bridges, and furniture.

Q7. Why are natural resources important?

Answer: Natural resources provide us with air, water, food, energy, shelter materials, and raw materials needed for life and human activities.

3. Renewable Resources

Q1. What are renewable resources?

Answer: Renewable resources are resources that are naturally replenished within a relatively short period or are continuously available.

Q2. Give examples of renewable resources.

Answer: Sunlight, wind, and flowing water are examples of renewable resources. Forests and soil can also be renewed, but they need proper care and time to recover.

Q3. Why is sunlight called a renewable resource?

Answer: Sunlight is continuously supplied by the Sun and is naturally available every day.

Q4. Is water a renewable resource?

Answer: Yes, water is renewed through the water cycle. However, usable freshwater is limited, so we must conserve it.

Q5. Are renewable resources unlimited in practical use?

Answer: Not always. Some renewable resources, such as forests and groundwater, can be depleted or damaged if they are used faster than they can recover.

4. Air and Its Composition (Page 154)

Q1. What is air?

Answer: Air is a mixture of gases present around the Earth.

Q2. What is the approximate composition of air?

Answer: Dry air contains approximately 78% nitrogen, 21% oxygen, and about 1% other gases, including argon and carbon dioxide. The amount of water vapour varies.

Q3. Which gas is present in the largest amount in air?

Answer: Nitrogen is present in the largest amount, about 78%.

Q4. Which gas is needed for breathing?

Answer: Oxygen is needed for breathing by humans and many other animals.

Q5. Why is air important for living things?

Answer: Air provides oxygen for respiration, carbon dioxide for photosynthesis, and other gases needed for natural processes.

Q6. What is wind?

Answer: Wind is moving air.

Q7. What is a breeze?

Answer: A breeze is a gentle movement of air.

Q8. What is a storm?

Answer: A storm is a period of disturbed weather that may include strong winds, rain, thunder, or other severe conditions.

Q9. Why do tree leaves move on a windy day?

Answer: Moving air pushes the leaves and makes them move.

Q10. What is humidity?

Answer: Humidity is the amount of water vapour present in the air.

Q11. How does humidity affect the climate of a place?

Answer: High humidity makes the air feel more uncomfortable and reduces the rate at which sweat evaporates from our skin. Humidity also affects cloud formation and rainfall.

Q12. What is global warming?

Answer: Global warming is the long-term rise in the average temperature of the Earth, mainly due to the increase in greenhouse gases caused by human activities.

5. Wind Power and Windmills (Page 155)

Q1. What is wind energy?

Answer: Wind energy is the energy obtained from moving air.

Q2. What is a windmill?

Answer: A windmill is a machine that uses wind to perform work, such as pumping water or grinding grain.

Q3. What is a wind turbine?

Answer: A wind turbine uses the energy of moving air to turn its blades and generate electricity.

Q4. How does a wind turbine generate electricity?

Answer: Wind turns the blades of the turbine. The rotating parts drive a generator, which produces electricity.

Q5. What is a wind farm?

Answer: A wind farm is an area where many wind turbines are installed to generate electricity.

Q6. Why are wind turbines installed in windy places?

Answer: Wind turbines need sufficient and steady wind to generate electricity efficiently.

Q7. Give two advantages of wind energy.

Answer:

1. Wind is a renewable source of energy.
2. Generating electricity from wind does not require burning fossil fuels during operation.

Q8. Give two limitations of wind energy.

Answer:

1. Wind speed changes, so electricity generation is not constant.
2. Wind turbines need suitable locations and may affect the local landscape and wildlife if poorly planned.

Think and Answer (Page 155)

Q. Could wind be used to generate electricity in your city? If so, suggest some likely locations. If not, what is the problem? Can the problem be solved?

Answer: Wind energy can be used if the city has places with strong and steady winds. Open fields, coastal areas, or suitable elevated locations may be considered after checking wind conditions. If winds are weak or buildings block the wind, large turbines may not be suitable. The city can consider solar energy or purchase wind-generated electricity from windy regions.

6. Water as a Natural Resource (Page 155)

Q1. Why is water an important natural resource?

Answer: Water is essential for drinking, cooking, washing, agriculture, industries, construction, and electricity generation.

Q2. How much of the Earth's surface is covered with water?

Answer: About 71% of the Earth's surface is covered with water.

Q3. Why is most of the Earth's water not suitable for drinking?

Answer: Most of the Earth's water is salty seawater. Only a small part is freshwater, and not all freshwater is easily available or safe to drink.

Q4. Where is freshwater found?

Answer: Freshwater is found in rivers, lakes, ponds, glaciers, snow, groundwater, and other freshwater sources.

Q5. What is groundwater?

Answer: Groundwater is water stored below the Earth's surface in soil and rocks.

Q6. Why should we conserve freshwater?

Answer: Freshwater is limited and essential for all living things. We must conserve it to meet present and future needs.

Q7. Name five daily activities that require water.

Answer: Drinking, cooking, bathing, washing clothes, and cleaning utensils require water.

Q8. How can we reduce water wastage in daily life?

Answer:

- Turn off taps when not in use.
- Repair leaking taps and pipes.
- Use a bucket instead of a running shower when possible.
- Reuse suitable water for gardening or cleaning.
- Collect rainwater.
- Use water carefully while washing vehicles and floors.

7. Rainwater Harvesting (Pages 155–156)

Q1. What is rainwater harvesting?

Answer: Rainwater harvesting is the process of collecting and storing rainwater for later use or for recharging groundwater.

Q2. What is rooftop rainwater harvesting?

Answer: Rooftop rainwater harvesting is a method in which rainwater falling on a roof is collected through pipes and stored in a tank or directed into the ground.

Q3. How does rooftop rainwater harvesting work?

Answer:

1. Rain falls on the roof.
2. Gutters and pipes collect the rainwater.
3. The water is filtered to remove dirt and other particles.
4. The filtered water is stored in a tank or directed underground to recharge groundwater.

Q4. What is surface run-off harvesting?

Answer: Surface run-off harvesting is the collection of rainwater flowing over the ground and directing it into pits, trenches, ponds, or recharge wells.

Q5. What is groundwater recharge?

Answer: Groundwater recharge is the process by which water enters the ground and replenishes underground water stores.

Q6. Why is rainwater harvesting important in cities?

Answer: Concrete roads and buildings prevent much rainwater from entering the ground. Rainwater harvesting helps collect water, reduce run-off, and recharge groundwater.

Q7. What are stepwells?

Answer: Stepwells are traditional structures with steps leading down to a water source. They were used to collect and access water, and some traditional systems also helped collect rainwater and groundwater.

Q8. Name two traditional water-harvesting structures mentioned in the chapter.

Answer: Baawadi in Rajasthan and Vav in Gujarat.

Q9. Can rainwater be used directly for drinking?

Answer: Rainwater should not be assumed safe to drink directly. It must be collected hygienically and treated appropriately before drinking.

Q10. Give three benefits of rainwater harvesting.

Answer:

1. It provides water for later use.
2. It helps recharge groundwater.
3. It can reduce surface run-off and waterlogging.

Think and Answer (Page 156)

Q. Name some daily activities where water is used and suggest methods to reduce water wastage.

Answer: Water is used for bathing, washing clothes, cooking, cleaning, and gardening. We can reduce wastage by closing taps, fixing leaks, using buckets, reusing suitable water, and collecting rainwater.

8. Energy from the Sun: Solar Energy (Page 156)

Q1. What is solar energy?

Answer: Solar energy is the energy we receive from the Sun in the form of light and heat.

Q2. Why is the Sun important for life on Earth?

Answer: The Sun provides light and heat. It supports photosynthesis, influences weather and the water cycle, and helps maintain conditions suitable for life.

Q3. How do plants use sunlight?

Answer: Plants use sunlight to make food through photosynthesis.

Q4. How are animals indirectly dependent on the Sun for food?

Answer: Plants use sunlight to make food. Animals eat plants or other animals, so food chains depend directly or indirectly on the Sun.

Q5. Name four uses of solar energy.

Answer:

1. Drying clothes.
2. Drying food grains.
3. Heating water.
4. Generating electricity.

Q6. Why is solar energy called renewable energy?

Answer: Solar energy is continuously supplied by the Sun and is naturally replenished.

Q7. What is one limitation of solar energy?

Answer: Solar energy generation depends on sunlight, so it is reduced at night and during cloudy weather.

Think and Answer (Page 156)

Q. What is the process by which plants make their own food?

Answer: The process is called photosynthesis. Green plants use sunlight, water, and carbon dioxide to make food and release oxygen.

9. Solar Panels and Solar Cells (Pages 156–157)

Q1. What is a solar cell?

Answer: A solar cell is a device that converts sunlight directly into electrical energy.

Q2. What is a solar panel?

Answer: A solar panel is a group of solar cells connected together to produce electricity.

Q3. What is a solar water heater?

Answer: A solar water heater uses energy from sunlight to heat water.

Q4. What is a solar cooker?

Answer: A solar cooker uses solar energy to cook food.

Q5. Give two advantages of solar energy.

Answer:

1. It is renewable.
2. It generates electricity without burning fuel during operation.

Q6. Why are solar panels useful in remote areas?

Answer: Solar panels can generate electricity in places where connecting to the electricity grid is difficult, provided there is sufficient sunlight and suitable equipment.

Q7. What is the difference between a solar cell and a solar panel?

Answer: A solar cell is a single electricity-generating unit. A solar panel contains many solar cells connected together.

10. Forests and Their Importance (Page 157)

Q1. What is a forest?

Answer: A forest is a large area covered mainly with trees and other vegetation, providing a habitat for many organisms.

Q2. Why are forests important?

Answer: Forests provide food and shelter for wildlife, help maintain biodiversity, absorb carbon dioxide, release oxygen through photosynthesis, protect soil, and support the water cycle.

Q3. What is biodiversity?

Answer: Biodiversity means the variety of living organisms found in an area or on Earth.

Q4. How do forests help animals?

Answer: Forests provide animals with food, water sources, shelter, and places to reproduce.

Q5. How do forests help humans?

Answer: Forests provide timber, fruits, medicines, fuelwood, fibres, and other useful products. They also support livelihoods and help protect the environment.

Q6. What is deforestation?

Answer: Deforestation is the large-scale cutting and clearing of forests for purposes such as agriculture, construction, and other human activities.

Q7. What are the harmful effects of deforestation?

Answer:

- Loss of wildlife habitats.
- Reduction in biodiversity.
- Soil erosion.
- Increased carbon dioxide in the atmosphere.
- Disturbance of the water cycle.
- Greater risk of floods and landslides in some areas.

Q8. What is forest conservation?

Answer: Forest conservation means protecting forests and using forest resources carefully so that they can continue to support life and regenerate.

Q9. What is Van Mahotsav?

Answer: Van Mahotsav is a tree-planting and forest-awareness campaign celebrated in India, generally during the first week of July.

Q10. Why should forests be managed responsibly?

Answer: Trees take many years to grow. Responsible management helps protect wildlife, maintain biodiversity, and ensure that forest resources remain available for future generations.

Think and Answer (Page 157)

Q. Imagine the Sun is not visible for a few days. What changes would there be in your daily activities?

Answer: Days would be darker, solar panels would generate less electricity, clothes and food would dry more slowly, and plants would receive less light for photosynthesis. The effects would depend on how long sunlight was unavailable.

11. Chipko Movement (Page 157)

Q1. What was the Chipko Movement?

Answer: The Chipko Movement was a forest-conservation movement in which local people hugged trees to prevent them from being cut down.

Q2. Where did the Chipko Movement begin?

Answer: The movement became well known in the early 1970s in the Uttarakhand region, then part of Uttar Pradesh.

Q3. Why did people participate in the Chipko Movement?

Answer: People wanted to protect forests that provided them with fuel, fodder, water security, and other essential resources.

Q4. What is the importance of the Chipko Movement?

Answer: It showed that community participation can help protect forests and natural resources.

Q5. How can students help conserve forests?

Answer: Students can plant and care for trees, avoid wasting paper, reuse notebooks when possible, and spread awareness about forest protection.

12. Non-renewable Resources (Page 157)

Q1. What are non-renewable resources?

Answer: Non-renewable resources are natural resources that are used up faster than they can be naturally replaced. Many take millions of years to form.

Q2. Give examples of non-renewable resources.

Answer: Coal, petroleum, natural gas, and many mineral resources are examples of non-renewable resources.

Q3. Why should non-renewable resources be used carefully?

Answer: They are limited and may take millions of years to form. Excessive use can lead to depletion and environmental damage.

Q4. What is the difference between renewable and non-renewable resources?

Renewable resources	Non-renewable resources
Naturally replenished on a relatively short timescale or continuously available.	Not replenished quickly enough to replace their use.
Examples: Sunlight and wind.	Examples: Coal, petroleum, and natural gas.
Some need careful management to remain available.	Need careful use because supplies are limited.

13. Soil (Pages 157–158)

Q1. What is soil?

Answer: Soil is the upper layer of the Earth's surface in which many plants grow. It contains mineral particles, organic matter, air, water, and living organisms.

Q2. Why is soil an important natural resource?

Answer: Soil supports plant growth, provides nutrients to plants, stores water, and is home to many organisms.

Q3. How is soil formed?

Answer: Soil forms slowly as rocks break down through weathering and mix with decomposed plant and animal matter over long periods.

Q4. What is weathering?

Answer: Weathering is the breaking down of rocks into smaller pieces by physical, chemical, or biological processes.

Q5. What is organic matter in soil?

Answer: Organic matter is material from living or once-living organisms, such as decayed leaves, roots, and animal remains.

Q6. What is soil fertility?

Answer: Soil fertility is the ability of soil to supply the nutrients needed for plant growth.

Q7. What are microorganisms?

Answer: Microorganisms are very small living organisms that usually cannot be seen clearly without a microscope.

Q8. How do soil organisms help the soil?

Answer: Many soil organisms break down dead plants and animals, release nutrients, and help improve soil structure.

Q9. Why does soil differ in colour and texture?

Answer: Soil colour and texture depend on the types and amounts of minerals, organic matter, and particle sizes present in it.

Q10. What are some uses of soil?

Answer: Soil is used for growing crops, supporting forests and grasslands, making bricks, and providing a base for buildings.

Q11. Why should we conserve soil?

Answer: Soil forms very slowly and is essential for food production and ecosystems. We should protect it from erosion, pollution, and degradation.

Activity: Observing Soil Samples (Page 158)

Q. How can you compare soil samples from different locations?

Answer: Collect samples separately from places such as a school garden, park, or field. Observe their colour, texture, and visible particles with a magnifying glass. Record the observations in a table.

Example observation table:

Location	Colour	Texture	Observation
School garden	Dark brown	Soft and crumbly	Some small roots and organic matter
Nearby field	Brown	Fine or slightly gritty	Small mineral particles
Sandy area	Light brown	Gritty	Many visible sand grains

These are sample observations. Actual results will depend on the soil collected.

14. Rocks (Page 158)

Q1. What are rocks?

Answer: Rocks are naturally occurring solid materials made up of one or more minerals.

Q2. Why are rocks important?

Answer: Rocks are important because they are used in construction, road building, making tools, and obtaining minerals.

Q3. Name three types of rocks mentioned in the chapter.

Answer: Marble, sandstone, and granite are examples of rocks.

Q4. What is slate used for?

Answer: Slate is used for roofing and other construction purposes.

Q5. What is laterite used for?

Answer: Laterite is used as a building material and for making bricks in some regions.

Q6. What is granite used for?

Answer: Granite is used in buildings, floors, kitchen countertops, monuments, and other construction work.

Q7. Why should rocks be used responsibly?

Answer: Rocks take thousands to millions of years to form. Excessive extraction can damage landscapes and ecosystems, so they should be used carefully.

Think and Answer (Page 158)

Q. What were the different ways in which rocks were used in ancient times?

Answer: In ancient times, rocks were used to make tools, weapons, grinding stones, shelters, monuments, and buildings. Some rocks were also used for making statues and ornaments.

15. Minerals (Page 159)

Q1. What are minerals?

Answer: Minerals are naturally occurring substances found in the Earth's crust. They have specific chemical compositions and characteristic properties.

Q2. Why are minerals important?

Answer: Minerals provide metals and other materials needed for buildings, vehicles, machines, electronic devices, and jewellery.

Q3. Name four metals obtained from minerals.

Answer: Iron, copper, gold, and aluminium are obtained from mineral resources.

Q4. What are the uses of iron?

Answer: Iron is used to make buildings, bridges, vehicles, tools, machines, and many other objects.

Q5. What are the uses of copper?

Answer: Copper is used in electrical wires, electronic devices, plumbing, and other equipment.

Q6. What are the uses of gold and silver?

Answer: Gold and silver are used in jewellery, ornaments, electronic components, and other specialised applications.

Q7. Why are minerals called non-renewable resources?

Answer: Mineral deposits take very long periods to form naturally and cannot be replaced quickly once extracted and used.

Q8. How can we conserve minerals?

Answer: We can conserve minerals by using materials efficiently, recycling metals, repairing objects, and reusing products instead of throwing them away.

Think and Answer (Page 159)

Q. What are the ways in which rocks and minerals are transported from one place to another?

Answer: Rocks and minerals are transported by trucks, tractors, trains, ships, and other vehicles. Large quantities may also be moved using conveyor belts.

16. Fossil Fuels (Pages 159–160)

Q1. What are fossil fuels?

Answer: Fossil fuels are fuels formed from the remains of ancient plants and other organisms over millions of years.

Q2. Name the three major fossil fuels.

Answer: Coal, petroleum, and natural gas.

Q3. How are fossil fuels formed?

Answer: The remains of ancient organisms were buried under layers of sediment. Over millions of years, heat, pressure, and other geological processes transformed this organic matter into fossil fuels.

Q4. Why are fossil fuels called non-renewable resources?

Answer: They take millions of years to form and are being used much faster than new supplies can form.

Q5. What is coal?

Answer: Coal is a solid fossil fuel formed mainly from ancient plant material. It is used in electricity generation and some industrial processes.

Q6. What is petroleum?

Answer: Petroleum, also called crude oil, is a naturally occurring liquid fossil fuel. It is processed in refineries to produce fuels such as petrol and diesel.

Q7. What is natural gas?

Answer: Natural gas is a gaseous fossil fuel made mainly of methane.

Q8. What is CNG?

Answer: CNG stands for Compressed Natural Gas. It is natural gas stored under high pressure and is used as a fuel in some vehicles.

Q9. What are the uses of fossil fuels?

Answer: Fossil fuels are used for transport, electricity generation, cooking, heating, and many industrial activities.

Q10. Why should fossil fuels be conserved?

Answer: Fossil fuels are limited resources. Burning them also releases carbon dioxide and other pollutants, so we should reduce unnecessary use.

Q11. What is the difference between coal, petroleum, and natural gas?

Coal	Petroleum	Natural gas
Solid fossil fuel.	Liquid fossil fuel in its natural form.	Gaseous fossil fuel.
Used in power generation and industries.	Processed to produce petrol, diesel, and other products.	Used for cooking, heating, electricity, and as a vehicle fuel.

Q12. Why is CNG often considered a cleaner vehicle fuel than petrol or diesel?

Answer: CNG generally produces less particulate matter and some other pollutants than petrol or diesel when burned in a suitable engine. However, it is still a fossil fuel and releases carbon dioxide.

Think and Answer (Page 159)

Q. What are the effects of burning fossil fuels on humans?

Answer: Burning fossil fuels releases air pollutants and greenhouse gases. Air pollution can cause breathing problems and other health issues, while greenhouse gases contribute to global warming and climate change.

Think and Answer (Page 159)

Q. How are rocks and minerals transported from one place to another?

Answer: They are transported by road, rail, ships, trucks, and conveyor belts, depending on the distance and quantity.

Try at Home (Page 160)

Q. How can you collect information about the vehicles used by your neighbours and the fuels they use?

Answer: Ask neighbours which vehicles they use and which fuels power them. Record the information in a table and present it to the class.

Vehicle	Fuel used
Motorcycle	Petrol or electricity, depending on the model
Car	Petrol, diesel, CNG, or electricity, depending on the model
Bus	Diesel, CNG, or electricity, depending on the model
Tractor	Usually diesel
Electric scooter	Electricity

This is a sample table, not a survey of actual neighbours.

17. Conservation of Natural Resources (Page 160)

Q1. What is conservation of natural resources?

Answer: Conservation of natural resources means protecting resources and using them wisely so that they remain available for present and future generations.

Q2. Why is conservation necessary?

Answer: Natural resources are limited or may be damaged by overuse. Conservation helps prevent depletion, pollution, and environmental harm.

Q3. What may happen if we continue to waste natural resources?

Answer: Resources may become scarce, ecosystems may be damaged, pollution may increase, and future generations may face shortages.

Q4. How can we conserve water?

Answer: We can repair leaking taps, avoid unnecessary water use, reuse suitable water, and collect rainwater.

Q5. How can we conserve forests?

Answer: We can plant and care for trees, reduce paper wastage, prevent illegal cutting, and use forest products responsibly.

Q6. How can we conserve fossil fuels?

Answer: We can use public transport, walk or cycle for short distances, share vehicles, save electricity, and use renewable energy where possible.

Q7. How can we conserve minerals?

Answer: We can recycle metals, repair products, reuse materials, and avoid unnecessary consumption.

18. Reduce, Reuse, and Recycle

Q1. What is the meaning of Reduce?

Answer: Reduce means using fewer resources and producing less waste.

Example: Avoid buying unnecessary items and reduce the use of disposable plastic products.

Q2. What is the meaning of Reuse?

Answer: Reuse means using an item again instead of throwing it away.

Example: Reuse jars for storage or use both sides of paper.

Q3. What is the meaning of Recycle?

Answer: Recycle means processing waste materials to make new useful products.

Example: Recycling paper, glass, aluminium, and suitable plastics.

Q4. Why are the three Rs important?

Answer: The three Rs help save resources, reduce waste, decrease pollution, and lower the need to extract new raw materials.

Q5. Which is generally better: reducing waste or recycling it?

Answer: Reducing waste is generally preferable because it avoids using resources and producing waste in the first place. Reuse and recycling are also important.

19. Sustainable Development (Page 160)

Q1. What is sustainable development?

Answer: Sustainable development means meeting the needs of the present generation without compromising the ability of future generations to meet their own needs.

Q2. Why is sustainable development important?

Answer: It helps balance human needs, economic development, and environmental protection.

Q3. What are the main goals of sustainable development in this chapter?

Answer:

- Prevent the depletion of natural resources.
- Reduce waste and excessive consumption.
- Recycle reusable materials.
- Prevent pollution.
- Protect the environment.
- Conserve forests and wildlife.

Q4. How can students support sustainable development?

Answer: Students can save water and electricity, plant trees, avoid wasting food, reuse books, segregate waste, and use reusable products.

Q5. How is conservation related to sustainable development?

Answer: Conservation protects natural resources, while sustainable development ensures that resources are used in a way that meets present needs without harming future generations.

Q6. Why should we protect wildlife and biodiversity?

Answer: Wildlife and biodiversity help maintain healthy ecosystems and support natural processes such as pollination, nutrient cycling, and food chains.

20. Important Application-Based Questions

Q1. A village faces water shortages during summer. Suggest three solutions.

Answer:

1. Collect and store rainwater.
2. Repair leaking water pipes and taps.
3. Use water efficiently and recharge groundwater where suitable.

Q2. A family wants to reduce its use of fossil fuels. What can it do?

Answer: It can use public transport, share vehicles, walk or cycle for short distances, save electricity, and use suitable renewable energy systems.

Q3. A forest is being cut down to make more land available. What problems may arise?

Answer: Wildlife may lose its habitat, biodiversity may decrease, soil erosion may increase, and the carbon balance and water cycle may be disturbed.

Q4. Why is it not wise to throw away an old metal object that can be repaired?

Answer: Repairing and reusing the object can reduce waste and save the minerals and energy needed to manufacture a new one.

Q5. Why are solar and wind energy useful alternatives to fossil fuels?

Answer: Solar and wind energy are renewable sources that can generate energy without burning fossil fuels during operation, reducing many forms of air pollution and greenhouse gas emissions.

Q6. Why should we not waste soil, rocks, and minerals even though they are found naturally?

Answer: They are valuable resources, and many form extremely slowly. Excessive extraction can damage land and ecosystems.

Q7. How does rainwater harvesting help a community?

Answer: It can provide water for later use, reduce surface run-off, and help recharge groundwater.

Q8. What is the difference between saving a resource and replacing it?

Answer: Saving a resource means reducing its use or waste. Replacing it means using another resource or material instead. For example, saving electricity reduces demand, while solar power can replace some electricity generated from fossil fuels.

21. Multiple-Choice Questions (MCQs)

Test Your Knowledge

0/15 answered

1. Which gas forms the largest part of air?

A. Oxygen
B. Nitrogen
C. Carbon dioxide
D. Hydrogen

2. What is moving air called?

A. Humidity
B. Wind
C. Steam
D. Smoke

3. Which of these is a renewable energy source?

A. Coal
B. Petroleum
C. Wind
D. Natural gas

4. What is the process of collecting and storing rainwater called?

A. Deforestation
B. Rainwater harvesting
C. Weathering
D. Recycling

5. Which device converts sunlight directly into electricity?

A. Windmill
B. Solar cell
C. Water pump
D. Coal furnace

6. Which movement helped protect forests by hugging trees?

A. Chipko Movement
B. Green Revolution
C. Water Movement
D. Swadeshi Movement

7. Which of these is a fossil fuel?

A. Sunlight
B. Wind
C. Coal
D. Flowing water

8. What is the upper layer of the Earth's surface in which plants grow called?

A. Cloud
B. Soil
C. Smoke
D. Petroleum

9. Which of these is a metal obtained from minerals?

A. Wood
B. Copper

C. Paper
D. Cotton

10. Which gas is a major greenhouse gas released by burning fossil fuels?

A. Carbon dioxide
B. Helium
C. Neon
D. Hydrogen

11. What does 'Reuse' mean?

A. Throwing an item away
B. Using an item again
C. Burning all waste
D. Buying more items

12. What does sustainable development aim to do?

A. Use all resources quickly
B. Meet present needs without harming future generations
C. Stop all human activities
D. Remove all forests

13. Which is an example of surface run-off harvesting?

A. Collecting flowing rainwater in pits
B. Burning coal
C. Cutting trees
D. Mining iron

14. Which is generally a non-renewable resource?

A. Sunlight
B. Wind
C. Coal
D. Rainfall

15. Which action helps conserve minerals?

A. Throwing away metal cans
B. Recycling metals
C. Wasting electronic devices
D. Extracting more than needed
Check Answers

22. Fill in the Blanks

1. Air is a mixture of gases.
2. The gas present in the largest amount in air is nitrogen.
3. Moving air is called wind.
4. The amount of water vapour in air is called humidity.
5. Energy obtained from the Sun is called solar energy.
6. Collecting and storing rainwater is called rainwater harvesting.
7. A group of solar cells forms a solar panel.
8. The large-scale cutting and clearing of forests is called deforestation.
9. The upper layer of the Earth's surface in which plants grow is called soil.
10. Coal, petroleum, and natural gas are fossil fuels.
11. The Chipko Movement helped protect forests.
12. Using an item again instead of throwing it away is called reuse.
13. Processing waste materials into new products is called recycling.
14. Meeting present needs without harming future generations is called sustainable development.
15. The breaking down of rocks into smaller pieces is called weathering.

23. True or False

1. Air is a mixture of gases. — True
2. Oxygen forms about 78% of air. — False
3. Wind energy is renewable. — True
4. Most of the Earth's water is freshwater. — False
5. Rainwater harvesting can help recharge groundwater. — True
6. Solar cells convert sunlight directly into electricity. — True
7. Deforestation can reduce biodiversity. — True
8. Soil forms very quickly in a few days. — False
9. Coal, petroleum, and natural gas are fossil fuels. — True
10. Fossil fuels are replenished as quickly as we use them. — False
11. Recycling metals can help conserve mineral resources. — True
12. Sustainable development considers the needs of future generations. — True
13. All renewable resources can be used without any limits or care. — False
14. Burning fossil fuels can increase carbon dioxide in the atmosphere. — True
15. The Chipko Movement was associated with protecting trees. — True

24. Quick Revision Table

Resource	Type	Main uses or importance
Air	Natural, renewable in broad terms	Breathing and natural processes
Water	Renewable through the water cycle, but limited locally	Drinking, farming, cleaning, industry
Sunlight	Renewable	Photosynthesis, heat, electricity

Wind	Renewable	Electricity and mechanical work
Forests	Renewable if responsibly managed	Wildlife habitat, timber, food, biodiversity
Soil	Slowly renewable and easily degraded	Agriculture and plant growth
Rocks	Generally treated as non-renewable on human timescales	Construction and raw materials
Minerals	Non-renewable on human timescales	Metals, electronics, buildings
Coal	Non-renewable	Electricity and industrial fuel
Petroleum	Non-renewable	Petrol, diesel, and other products
Natural gas	Non-renewable	Cooking, electricity, and vehicle fuel

25. Important Exam Definitions

- Natural resource: A useful material or substance obtained from nature.
- Renewable resource: A resource that is naturally replenished on a relatively short timescale or is continuously available.
- Non-renewable resource: A resource that cannot be replaced as quickly as it is used.
- Wind energy: Energy obtained from moving air.
- Rainwater harvesting: Collecting and storing rainwater for later use or groundwater recharge.
- Solar energy: Energy obtained from the Sun.
- Deforestation: Large-scale clearing of forests.
- Biodiversity: The variety of living organisms in an area.
- Soil fertility: The ability of soil to supply nutrients needed by plants.
- Weathering: The breaking down of rocks through natural processes.
- Fossil fuels: Fuels formed from ancient organic matter over millions of years.
- Conservation: Protection and careful use of natural resources.
- Sustainable development: Meeting present needs without compromising the ability of future generations to meet their own needs.

Exam tip: Focus especially on the differences between renewable and non-renewable resources, the composition of air, wind and solar energy, rainwater harvesting, the Chipko Movement, soil and minerals, fossil fuels, and the three Rs. The uploaded PDF contains textbook pages 153–160; it does not include a separate end-of-chapter exercise page. The answers above cover the questions visible in these pages and add topic-wise practice questions.

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